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7.1 Introduction

Control systems, a class of embedded systems, focus on tracking the reference input that is provided to the system. Initially the reference input is set and the output is more likely to track the same input regardless of the different external factors involved. The tracking can get difficult with the presence of disturbances. However, the system must be able to adjust to external factors for optimum performance. The objective of a control system is to track the reference output. The following figures represent good tracking and bad tracking respectively.

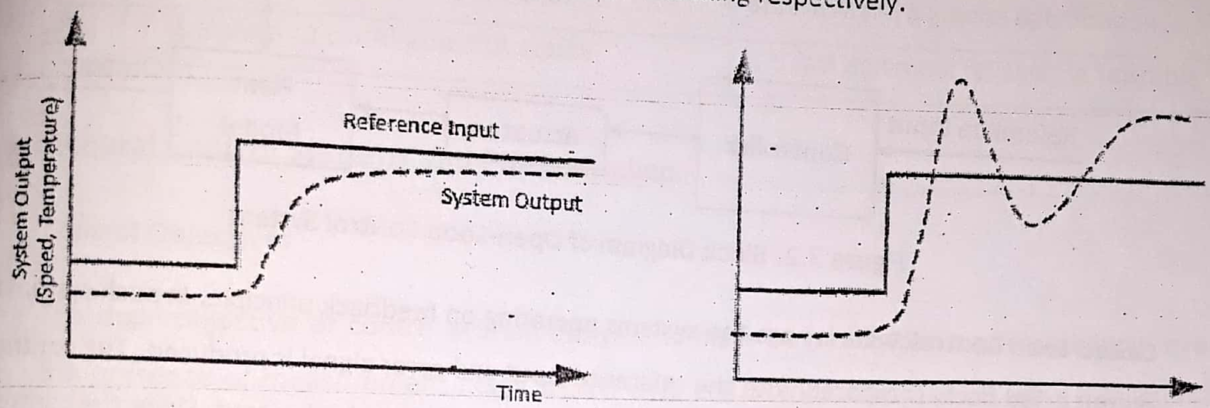

Figure 7.1: Good Tracking and Bad Tracking

7.2 Open-Loop and Closed-Loop Control Systems Overview

Open-Loop Control Systems are those systems in which the output has no influence on the control action of the input signal. It is also referred as feed-forward system or non-feedback system since the output is not fed back for comparison with the reference input. Also the controller is not aware about the tracking of reference input, so optimization is not possible. These systems are best utilized in case of predictable systems whose model is accurate and disturbance effect is minimal. In general, the open-loop control systems consist of following:

- Plant, which is also referred as a process, is the physical system to be controlled. Automobiles, fan, heater, disk etc are few examples.
- Output is the aspect or attribute of the physical system that we are about to control. Speed, temperature can be taken as examples.
- Reference input is the desired value that is required to be observed as an output of the physical system. Desired speed, temperature set by the user represents a reference input.
- Actuator is the device that is used to control the input to the plant. Motor can be taken as an example of an actuator.

- **Controller** is the main processing part of the system which computes the input to the plant such that desired output is achieved based on given reference input.
- **Disturbance** is an undesirable input to the system that may cause the output to deviate from the desired reference input.

The general block diagram of Open-Loop Control Systems is shown in the figure below.

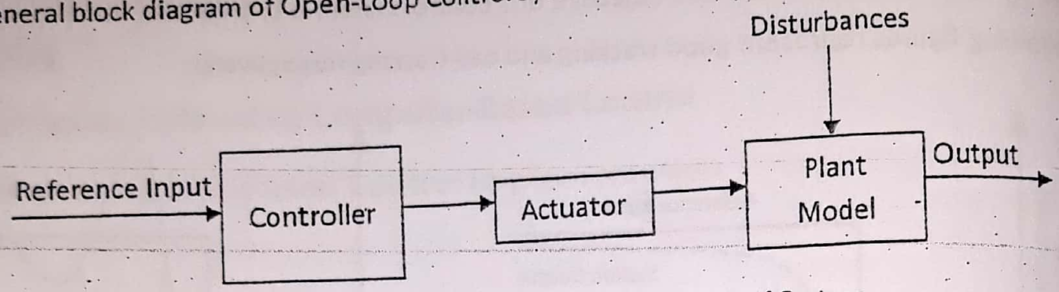

Figure 7.2: Block Diagram of Open-Loop Control System

Closed-Loop Control Systems are the systems operating on feedback principle. In such system the output is fed back, compared with the reference input and error signal is produced. The controller processes the error signal and reduces the error to obtain the desired output. Since the controller is aware about the output variations, optimization can be done and optimum performance can be obtained by minimizing the error. Apart from the plant, output, reference, controller, actuator, and disturbances, closed-loop control system contains additional components as sensor and error detector.

- **Sensor** is used to sense the output of the system and is fed to the input where error is calculated.
- **Error Detector** determines the error being produced in the system. Error is calculated by determining the difference between the output of the system and the reference input.

The general block diagram of Closed-Loop Control Systems is shown in the figure below.

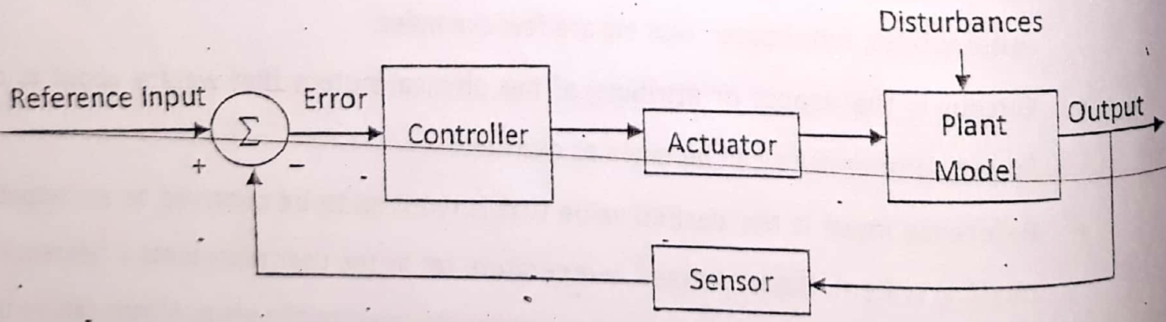

Figure 7.3: Block Diagram of Closed-Loop Control System

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Comparison of Open-Loop and Closed Loop Control Systems

SN	Open-Loop Control System	Closed-Loop Control System
1.	Feed Forward System: Output is not fed back	Feed Back System: Output is fed back and compared with input
2.	It is simple and economical.	It is complex and expensive
3.	Good calibration can lead to good accuracy but optimization is not possible	Feedback principle reduces error, increases accuracy and supports optimization
4.	It is slow and unreliable but stable	It is fast and more reliable but unstable

7.3 General Control Systems and PID Controllers

Control Objectives

The main objective of control system design is to make output track the reference input even in the presence of measurement noise, model error and disturbances. The objective fulfillment can be analyzed and assessed through various metrics.

- **Stability**: For the system to be stable, all variables in the system remain bounded
- **Performance**: It describes how well the output is tracking the change in the reference input. The various aspects of performance is shown in the figure below:

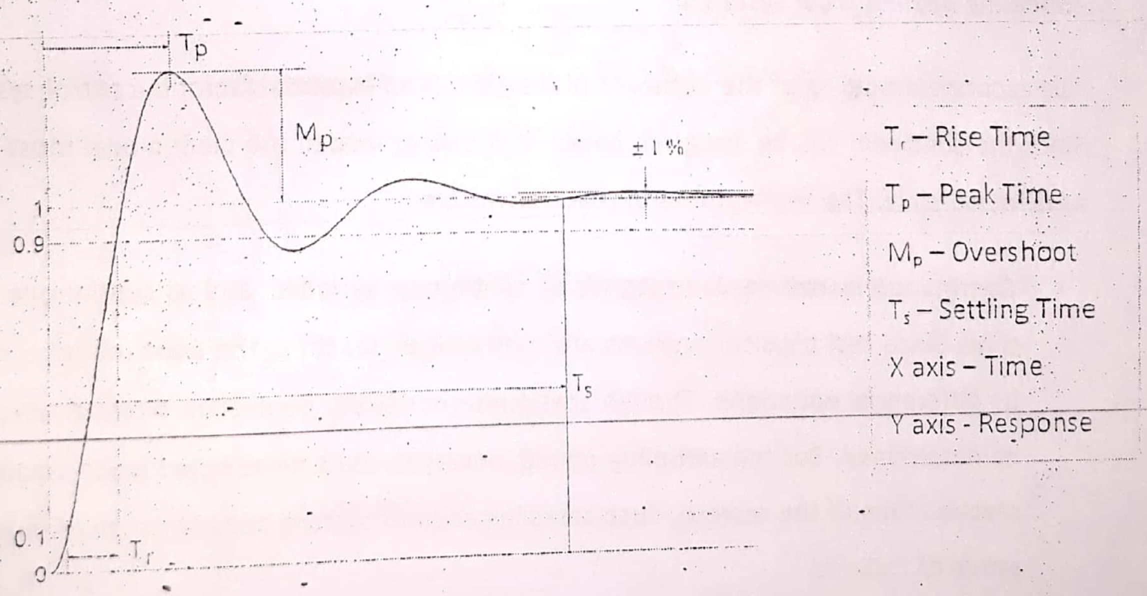

Figure 7.4: Aspect of performance metrics in Control system response.

The different aspects of performance are discussed below:

- **Rise Time (T_r)** is the time required to change from 10% to 90% of its final value. It is a measure of the ability of a system to fast input signals.
- **Peak Time (T_p)** is the time required to reach the first peak of the response.
- **Overshoot (M_p)** refers to an output exceeding its final, steady-state value. It is the percentage amount by which the peak of the response exceeds the final value.
- **Settling Time (T_s)** is the time required for the system to settle down to within 1% of its final value.

- **Disturbance rejection:** Disturbances are the undesired effects which cannot be eliminated but its impact can be minimized.
- **Robustness:** The system to be designed must be able to tolerate the modeling error of the plant. The stability and performance of the system should not be significantly affected by the presence of model errors.

Transient Response and Steady State Response of Control System

Transient response occurs just after the system starts and when any undesired conditions occur. The system's response during the settling time is transient response. Whereas the Steady state occurs after the system becomes settled. Steady State Error is defined as the difference between the actual output and the desired output when system reaches steady state.

Modeling Real Physical Systems

The accurate modeling of the behavior of the plant is an essential factor in control system design. Since the controller will be designed based on the plant model, the plant model must be accurate as far as possible. The key features of real systems are:

- **Continuous in nature:** It responds as continuous variables and as continuous function of time. Since real physical systems are continuously reacting, the plant model is represented by differential equations. Though continuous in nature, equivalent discrete time model can be determined. But the sampling period, however, must be selected much smaller than the reaction time of the system. Such sampling ensures system does not change much between sampling instants.

- **Complexity:** It is much more complex than any model we generally assume in our design. Our model may not include nonlinear effects, all system states, or all system interactions. Generally assumed model is a linear model which is sufficient when the variables of the model have a small operating range.

Controller Design

Proportional Control

A Controller that multiplies the tracking error by a constant is referred as proportional control. The form of proportional control is:

$u(t) = P * e(t)$

Where, $u(t)$ is the output of the controller, P is the proportional Constant, $e(t)$ is the measured error and is the difference between reference input and output of the system.

Proportional Constant affects transient response, steady state tracking error and disturbance rejection. High value of proportional constant can cause system to become unstable by resulting in high overshoot and oscillation, whereas low value of P will cause the system to be less response or less sensitive, since rise time will be high for low value of P. Also the steady state error will be high for low value of P. The following figure shows the response of an arbitrary control system for different values of P.

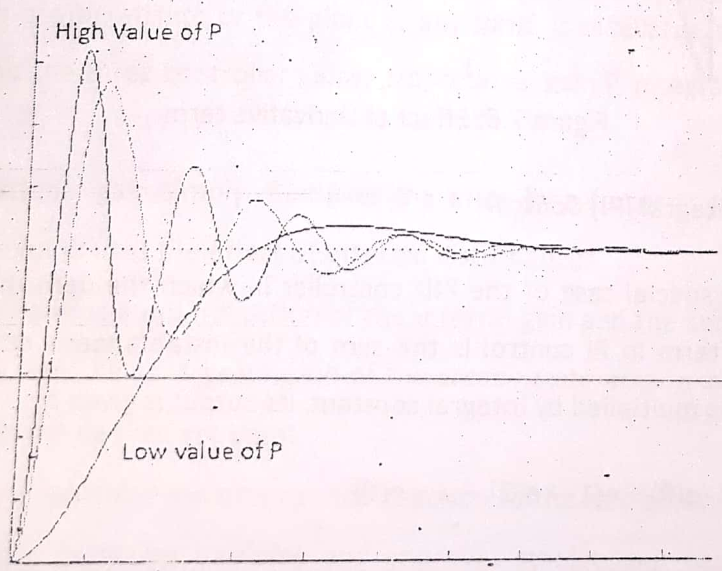

Figure 7.5: Response of system for different values of proportional constant

Proportional and Derivative (PD) Control

Derivative action predicts system behavior and improves settling time and stability of the system. Derivative term allows the transient response to be optimized without affecting the steady state response and disturbance rejection characteristic. Hence, transient response and the steady state error independently can be adjusted by using appropriate values of P and D in PD controller. The form of PD control is:

u(t) = P * e(t) + D * (e(t) – e(t-1))

Characteristics of PD control:

- Rise time reduces, improves damping, overshoot reduces, response is stable.

The following figure shows the response of an arbitrary system for PD control action.

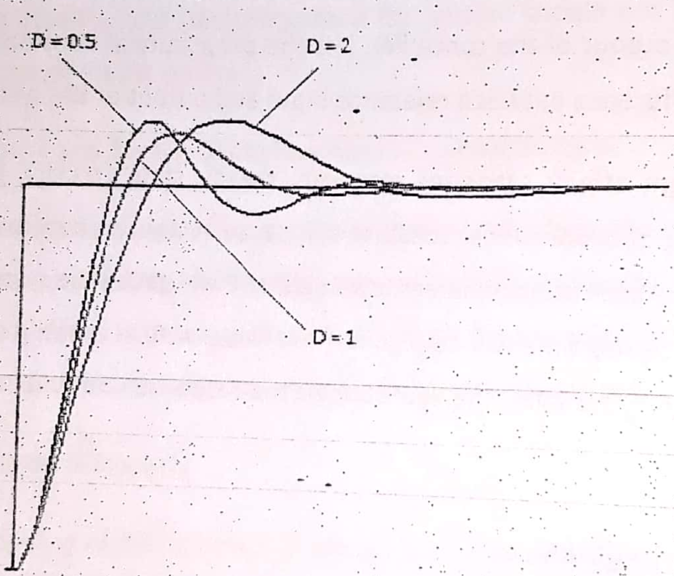

Figure 7.6: Effect of Derivative term

Proportional and Integral (PI) Control

A PI controller is a special case of the PID controller in which the derivative of the error is not used. The integral term in PI control is the sum of the instantaneous error over time and the accumulated error is multiplied by integral constant. Its output is given by

u(t) = P * e(t) + I * (e(0) + e(1) + e(2) + ... + e(t))

PI controller is used to eliminate the steady state error resulting from P controller. However, it has undesirable impact on speed and stability of the system.

Characteristics of PI control

- Steady state accuracy improves, rise time increases, response is oscillatory

The following figure shows effect of different values of integral constant in the response of an arbitrary system for PD control action.

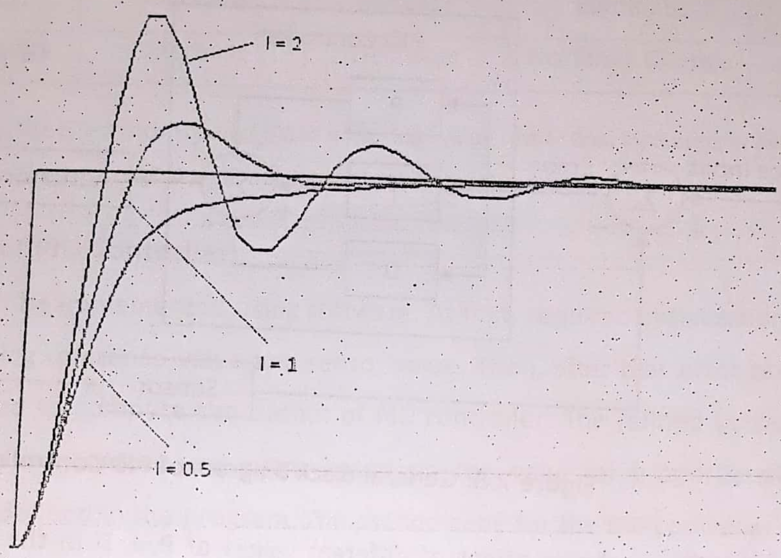

Figure 7.7: Effect of Integral Term in system response

Proportional Integral and Derivative (PID) Control

PID controller is a feedback controller that helps to attain a set point irrespective of disturbances or any variation in characteristics of the plant of any form. It calculates its output based on the measured error and the three controller gains; proportional gain P, integral gain K, and derivative gain D.

- The proportional gain simply multiplies the error by a factor P. It reduces steady state errors while minimizes the effect of external disturbances.
- The integral term is a multiplication of the integral gain and the sum of the recent errors. The integral term helps in getting rid of the steady state error and causes the system to catch up with the desired set point.
- The derivative controller determines the reaction to the rate of which the error has been changing and it increases damping and improves stability but has almost no effect on steady state error.

Its output is given by

u(t) = P * e(t) + I * (e(0) + e(1) + e(2) + ... + e(t)) + D * ((e(1) – e(0))+(e(2) – e(1))+... + (e(t) – e(t-1)))

The general block diagram of PID controller is shown in the figure below:

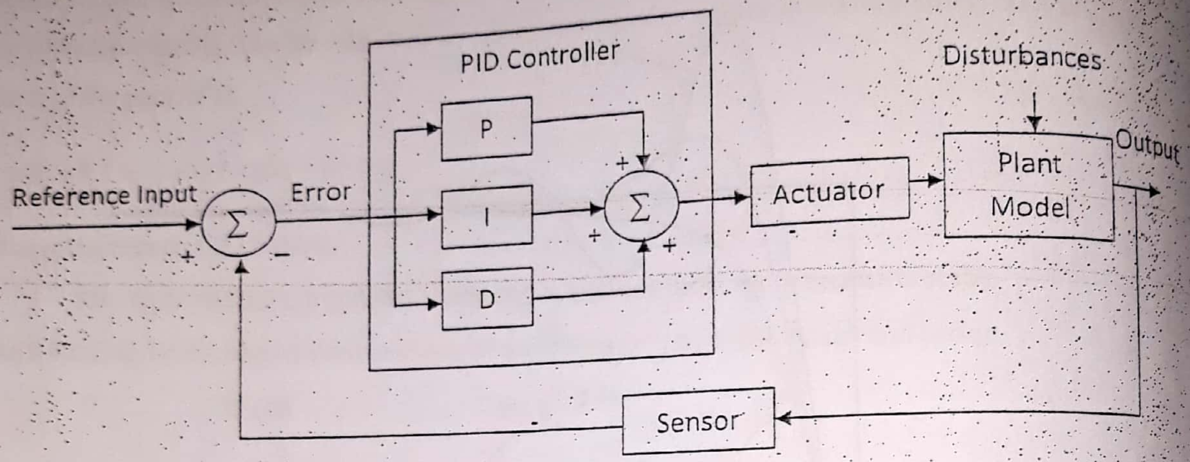

Figure 7.8: General Block Diagram of PID Controller

The following figure shows effect of different values of P, I, D in the response of an arbitrary system for PID control action.

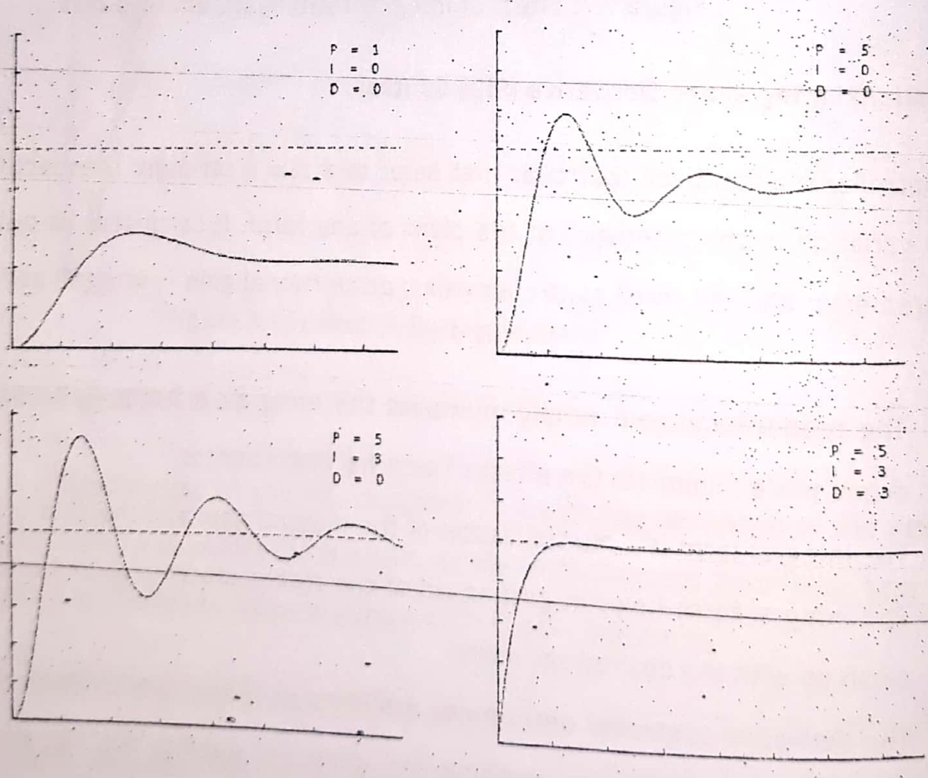

Figure 7.9: Effects of different values of P, I, D in the response of an arbitrary system

Summary of PID control action

Type	Rise Time	Maximum Overshoot	Settling Time	Steady-state error	Stability
P	Decrease	Increase	Small Change	Decrease	Degrade
I	Decrease	Increase	Increase	Eliminate	Degrade
D	Small Change	Decrease	Decrease	No/Small Change	Improve

(*Note: In above table, the effect is considered based on optimal value rather than increasing or decreasing the value of Proportional, Integral and Derivative constant)

7.4 Software Coding of PID Controller

A PID controller can be implemented using software. At first, required initialization is done which is followed by reading reference value and sensor value. Then, after that error can be calculated which further is used to compute the output of PID controller. The refined output is fed to the actuator which in turn controls the plant based on the value of proportional, integral and derivative constant defined in the program. The pseudo code for the PID controller can be written as:

- → Set values for Pgain, Igain, Dgain
- → Initialize prior_error = 0 and integral = 0
- → Repeat following steps
  - sensorValue = getValueFromSensor()
  - refValue = getReferenceValue()
  - error = refValue – sensorValue
  - integral = integral + error*iterationTime
  - derivative = (error – prior_error)/iterationTime
  - output = Pgain * error + Igain * integral + Dgain * derivative
  - setActuator(output)
  - prior_error = error
  - wait(iterationTime)

7.5 PID Tuning

PID tuning is the adjustment of its control parameters to the optimum values for the desired control response. Quantitative analysis can be used to determine the values of P, I, and D. However, quantitative analysis is not necessary when safety and cost of using plant is not a concern. There are various methods for PID tuning, one of which is ad hoc tuning process. The steps for ad hoc tuning process are

- Start with small value of P gain, D and I gains as 0
- Increase value of D gain until oscillation is seen, and then D gain is decremented by a factor of 2 to 4.
- Then, increase value of P gain until oscillation or excessive overshoot is observed, and then P gain is reduced by a factor of 2 to 4.
- Next, increase the value of I gain and reduce it slightly when oscillation or excessive overshoot is seen.
- Above steps are repeated until satisfactory performance is achieved.

7.6 Practical Issues Related to Computer-Based Control

The various practical issues related to computer-based control are explained in the following paragraphs.

a. Quantization and Overflow Error

Quantization error occurs when machine number is altered to fit the constraints of the computer memory.

- Case I: when arithmetic results require more precisions than original values. For example, in operation 0.50 x 0.25 = 0.125, the final result requires more precision.
- Case II: when analog signals from sensors are quantized by analog to digital converter it can create quantization error. In quantization process limited set of discrete values are defined and if the signal or value from the sensors doesn't match the defined quantized discrete values then rounding or truncation will occur which results in quantization error. For example: When 4 levels are defined between -1.5 and 1.5 as -1.5, -0.75, 0, 0.75 and 1.5 then the value 1.3 will be taken as 1.5.

Overflow error occurs when the system attempts to operate on or results a number that does not lie within the defined range of the system. For example, let us consider a case of signed binary numbers where five bits are used to represent the magnitude while sixth or MSB is used to represent sign. Using such representation, when two binary numbers 010010 (+18) and 010101 (+21) are added then it results in 100111 which is (-25) rather than (+39).

Since the first bit is used for sign representation, the undesirable output resulted due to overflow error. The situation can get more complex if we consider multiplication operation and floating point numbers.

b. Aliasing

Aliasing is the consequence of improper sampling process. It arises when a signal is discretely sampled at a rate that is insufficient to capture the changes in the signal. In simple term, aliasing causes the reconstructed signal to be different from original signal. It causes different signals to become indistinguishable. Let us consider an example in which the sampling is done at a period of 0.4 second which results a sampling frequency of 2.5 Hz. Then the following signals will be indistinguishable

$y(t) = 1.0 * \sin(6\pi t)$, frequency 3 Hz

$y(t) = 1.0 * \sin(\pi t)$, frequency 0.5 Hz

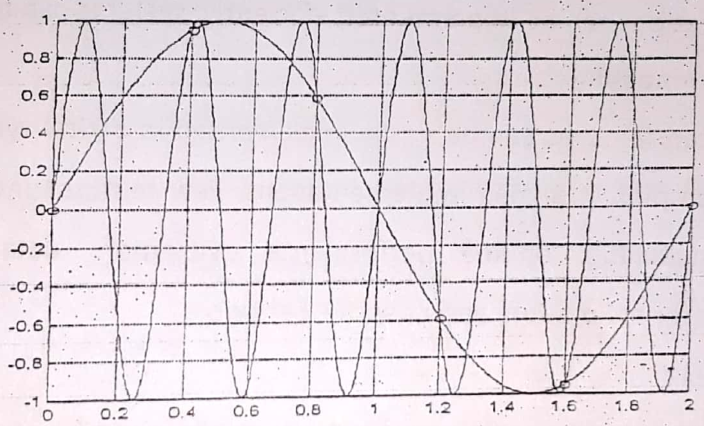

Figure 7.10: Aliasing Illustration

For a sampling rate of 2.5 Hz, sine wave with frequency of 0.5 Hz is indistinguishable from sine waves at 3 Hz, 5.5 Hz, 8 Hz and so on. Also, it can only correctly sample signal below Nyquist Frequency, which is half the value of sampling rate.

c. Computation Delay

Delay results in control signal being applied later than desired time. Computation delay is the attribute of many digital systems but too much delay results in performance degradation. The effect of delay can be accurately analyzed and we need to characterize implementation delay to ensure its effect is negligible in system's performance. Synchronous Design makes hardware delay to be characterized easily. Software delay, however, is harder to predict. So, code should be organized carefully to make delay predictable. Also code can be written with predictable timing behavior, such that the effect of delay can be minimized to acceptable level.

7.7 Benefits of Computer-Based Control Implementation

The following are the benefits of computer-based control implementation.

a. **Repeatability**

Analog systems are more prone to aging, temperature and manufacturing tolerance effects which cause results to vary with time. However, the digital systems can produce identical results for longer time

b. **Stability**

Since digital systems are less prone to different sorts of degradations and optimizations can be implemented efficiently, systems can become more stable.

c. **Programmability**

Advanced features can be easily implemented in digital systems but that would be very complex in analog implementations. Few features include: control mode and gain switching, on-line performance evaluation, data storage, performance parameter estimation, and adaptive behavior.

d. **Flexibility**

Computer based control can be easily re-configured based on requirement which allows periodic upgrade and enhancement of the system. It permits modification of the sequencing and control procedures for different products and for frequent change in product specifications.